

CSS3 Cheat Sheet

Box model, Flexbox, Grid, positioning & responsive design — the styling rules you'll reach for daily.

Flexbox

Grid

Box Model

Responsive

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1 Selectors at a Glance

Selector	Matches
.class	Every element with that class
#id	The single element with that id
elem1 elem2	elem2 nested anywhere inside elem1 (descendant)
elem1 > elem2	elem2 that is a direct child of elem1
a:hover	Element in a particular state (hover, focus, active)
li:first-child / nth-child(2)	Position-based matching inside a parent
::before / ::after	Inserts generated content before/after an element

2 The Box Model

```
.box {
  width: 200px;
  padding: 16px; /* inner space */
  border: 1px solid #ccc;
  margin: 12px; /* outer space */
  box-sizing: border-box;
}
```

📦 Order, outside-in

- **Content** → text/children
- **Padding** → space inside the border
- **Border** → the visible edge
- **Margin** → space outside, between elements

`box-sizing: border-box` keeps width/height predictable — set it globally.

3 Flexbox — One-Direction Layouts

```
.row {
  display: flex;
  flex-direction: row; /* or column */
  justify-content: center; /* main axis */
  align-items: center; /* cross axis */
  gap: 12px;
  flex-wrap: wrap;
}
```

🎯 justify-content values

`flex-start` · `center` · `space-between` · `space-around` · `space-evenly`

✏️ child controls

`flex: 1` grows to fill space · `flex-shrink: 0` prevents shrinking

4 CSS Grid — Two-Direction Layouts

```
.grid {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  grid-template-rows: auto 1fr;
  gap: 16px;
}
.item { grid-column: span 2; }
```

🧩 When to choose which

Flexbox → navbars, button groups, single row/column alignment.

Grid → full page layouts, card galleries, dashboards (rows + columns together).

5 Positioning

static (default)

relative

absolute

fixed

sticky

📌 **Rule of thumb:** `absolute` positions relative to the nearest ancestor with `position: relative` — set that ancestor first.

6 Units Cheat Table

Unit	Best for
px	Fixed, pixel-precise sizing (borders, small icons)
%	Size relative to the parent element
em / rem	Relative to font-size (rem = root font-size) — scalable type
vw / vh	Relative to viewport width/height — full-screen sections

7 Responsive Design

```
/* mobile-first breakpoint */
.card { width: 100%; }

@media (min-width: 768px) {
  .card { width: 50%; }
}
```



Common breakpoints

576px sm · 768px md · 992px lg · 1200px xl

Design mobile-first, then add `min-width` media queries as screens grow.

8 Transitions & Simple Animation

```
.btn {
  transition: all 0.2s ease-in-out;
}
.btn:hover { transform: scale(1.05); }

@keyframes fadeIn {
  from { opacity: 0; } to { opacity: 1; }
}
.card { animation: fadeIn 0.4s ease-in; }
```

9 Colors, Backgrounds & Typography



Color formats

#hex · rgb(0,0,0) · rgba(0,0,0,0.5) · hsl(210,90%,55%)



Backgrounds

background: url(img.jpg) center/cover no-repeat;



Typography

- font-family, font-size, font-weight
- line-height for readable spacing
- text-align, letter-spacing

10 Common Beginner Mistakes



Forgetting `box-sizing: border-box`, using `!important` to "fix" specificity instead of understanding it, and centering with `margin: auto` without first setting a fixed `width`.